

Kourai Khryseai: Transparent Human-on-the-Loop Multi-Agent Software Development

AJ Barea • Rochester Institute of Technology • Advisor: Dr. Leon Reznik
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— ABSTRACT

Multi-agent coding systems are increasingly capable, but many still hide coordination decisions from users. **Kourai Khryseai** treats multi-agent work as a *supervised collaboration problem*: an orchestrator routes requests to specialists while streaming intermediate status and pausing on ambiguity — making coordination *legible* rather than maximizing benchmarks.

— INTRODUCTION & RELATED WORK

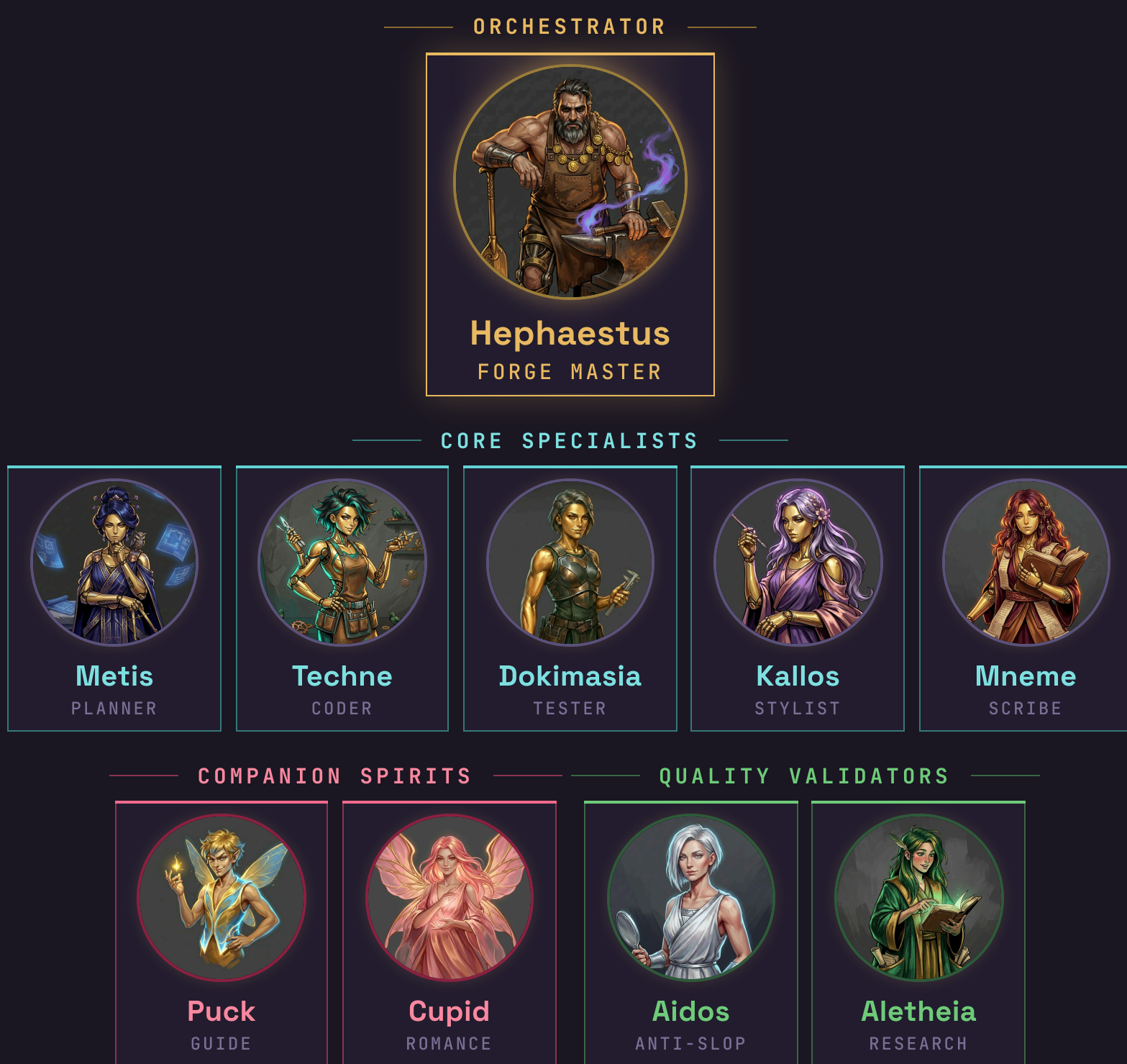
Benchmarks like SWE-bench [3] and platforms like OpenHands [5] push issue-resolution forward. Magentic-One [2] explores multi-agent orchestration; HULA [4] puts humans directly in the loop; AgentTrace [1] focuses on observability. **Kourai sits at this intersection**: explicit orchestration, streamed intermediate state, and interactive supervision.

SYSTEM	MULTI-AGENT	STREAMED	HOTL
SWE-bench [3]	—	—	—
OpenHands [5]	✓	partial	—
Magentic-One [2]	✓	—	—
HULA [4]	—	✓	HITL
AgentTrace [1]	—	✓	—
Kourai	✓	✓	HOTL

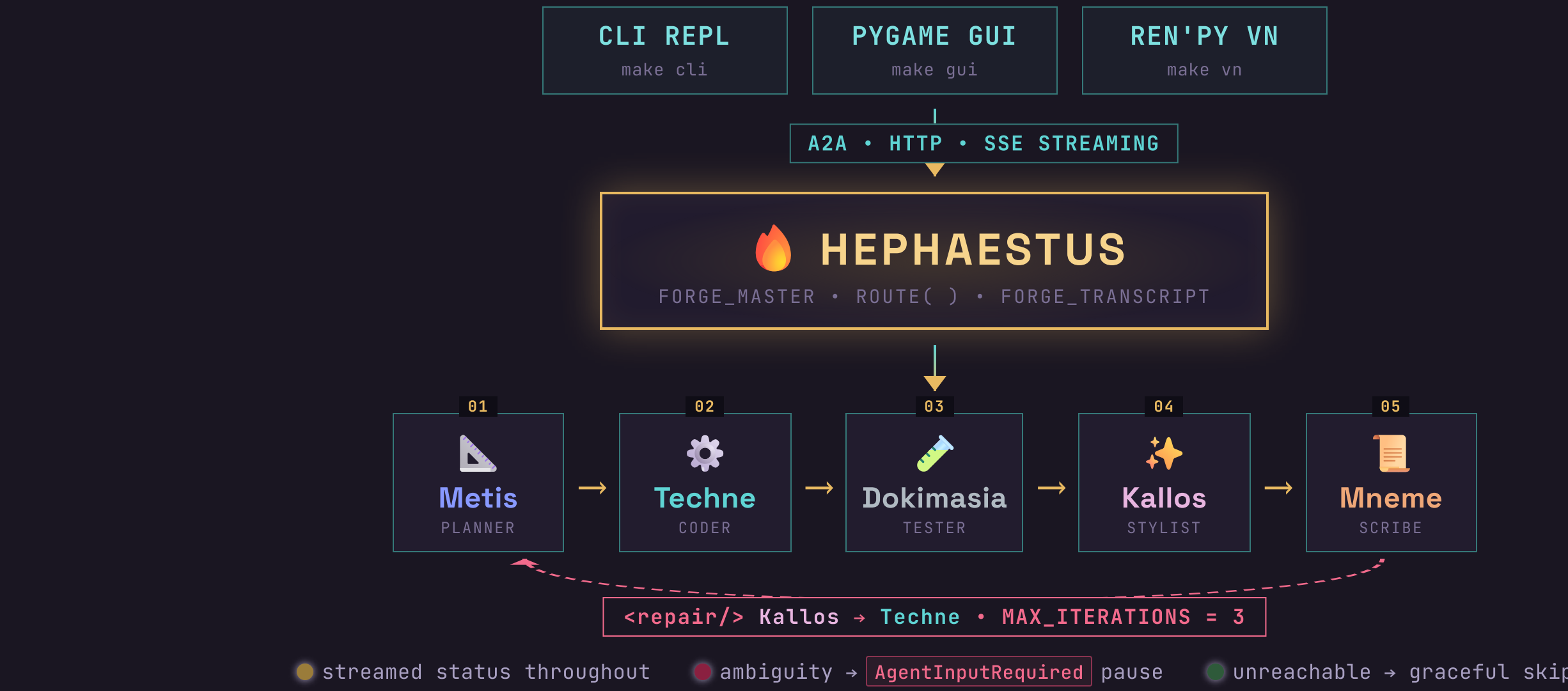
— HOTL VS HITL — A DELIBERATE SHIFT

HITL	HOTL
user <i>authors</i> each step	user <i>supervises</i> & redirects
Specialists act; the user steps in when a decision matters or quality drifts.	

— THE FORGE PARTY — 10 AGENTS



— SYSTEM DESIGN — ORCHESTRATED SPECIALIST PIPELINE



ROUTING BY INTENT — 7 PIPELINE TEMPLATES

implement x	metis → techne → dokimasia → kallos → mneme
fix bug x	techne → dokimasia → kallos → mneme
add tests for x	dokimasia → kallos → mneme
clean up x	kallos → mneme
commit prep	mneme
plan x	metis
lint/format x	kallos

— FORGE MASTER NARRATES — IN-CHARACTER HANDOFFS

Each handoff emits an in-character line from **Hephaestus**, making orchestration legible as dialogue.

→ METIS	"Draw up the plans. And no improvising."	→ TECHNE	"Write something worthy of my forge."
→ DOKIMASIA	"Find every flaw. Leave nothing."	→ KALLOS	"Go make it pretty or whatever you do."
→ MNEME	"Write it down. The FACTS, not your opinions."		

— STREAMED HOTL — CLI SAMPLE

```
> implement CSV export with tests
● Hephaestus: Metis! Draw up the plans.
↳ Metis: analyzing...
  ↳ streaming for >100MB files? (y/n)
> yes, chunked I/O
  ↳ Metis: spec ready ✓
  ↳ Techne: writing files... ✓
  ↳ Dokimasia: 12/12 passing ✓
  ↳ Kallos: no issues ✓
  ↳ Mneme: commit drafted ✓
```

— ONE BACKEND • THREE HOSTS

Same orchestration logic under all three. Interface varies — terminal REPL, pygame chat with TTS voices, illustrated visual novel — while backend logic stays fixed. A platform for studies of UI-agnostic agent behavior.

CLI REPL

status stream • keyboard checkpoint

Kourai Khryseai — Golden Maidens

charged in fire, refined by hand

Enter send • Ctrl+V paste • Alt+V attach image
/help commands • /q quit

Connected to Hephaestus v0.9.2 • 10 agents online • forge warm
Model: claude-opus-4.7 • demo mode (no network, no LLM)

> Implement CSV export with tests for the events module

● Hephaestus: "Metis! Draw up the plans. And no improvising."

↳ Metis: analyzing events module...

↳ Found 3 models with datetime fields
↳ nested relationships in Attendance and Session

↳ Metis: "One decision before I draft the spec."

↳ "Should CSV export stream chunked I/O for large files?"
events.json has ~40k rows in prod. Loading all at once would spike memory. Streaming stays constant-memory.

[y] yes, stream chunked [n] no, load in memory
[?] explain trade-offs [C] cancel

> ||

PYGAME GUI

portrait chat • free-form input

METIS
Structure is beautiful. I am both.

KOURAI KHRYSEAI — Golden Maidens
HEPHAESTUS — The Forge Master
The forge is hot. What are we building?

connected 20:00:01

implement CSV export with tests for the events module 20:00:01

Metis — show me what that golden brain of yours can do. 20:00:01

METIS — The Architect
found 3 models with datetime fields; nested relationships in Attendance and Session. 20:00:01

METIS — The Architect
"One decision before I draft the spec." 20:00:01

METIS — The Architect
"Should CSV export stream chunked I/O for large files? events.json has ~40k rows in production — loading all at once would spike memory. Streaming stays constant-memory but adds ~3ms/row of overhead." 20:00:01

(Commit) (Link) (Test)

> [Metis is waiting for input] Type your answer...

REN'PY VN — HANDOFF

routing spoken aloud • agents named inline

HEPHAESTUS
ARCHITECT OF THE FORGE

Metis — Draw up the plans. And no improvising.

REN'PY VN — CHECKPOINT

ambiguity becomes a menu choice

HEPHAESTUS
ARCHITECT OF THE FORGE

METIS
ARCHITECT OF INTENT

Yes, stream chunked.
No, load in memory.
Explain the trade-offs.

Should CSV export stream chunked I/O for large files? events.json has ~40k rows in production — loading all at once would spike memory. Streaming stays constant-memory but adds ~3ms/row of overhead.

— ARTIFACT EVALUATION — TEST PROPERTIES

PROPERTY	TEST FOCUS	✓
Pipeline ordering	end-to-end canonical route	✓
Forge Transcript accumulation	shared context per specialist	✓
Clarification loops	AgentInputRequired pause/resume	✓
Bounded Kallos!Techne repair	MAX_ITERATIONS = 3	✓
Graceful degradation	"Skipped (unreachable)"	✓

100
UNIT TEST FILES

9
INTEGRATION FILES

~2,000
TEST FUNCTIONS

— CONCLUSION

"Transparency is a systems property, not just a UX choice."

Explicit specialization improves **modularity** and **debuggability**. Streamed coordination and HOTL checkpoints make agent behavior easier to *inspect* and *redirect* — a platform for future studies of agent memory, orchestration, debugging, and user trust.

— REFERENCES

- [1] A. AlSayyad, K. Y. Huang, R. Pal. *AgentTrace: A structured logging framework for agent system observability*. arXiv:2602.10133, 2026.
- [2] A. Fourney et al. *Magentic-One: A generalist multi-agent system for solving complex tasks*. arXiv:2411.04468, 2024.
- [3] C. E. Jimenez et al. *SWE-bench: Can language models resolve real-world GitHub issues?* ICLR 2024.
- [4] W. Takerngsaksiri et al. *Human-in-the-loop software development agents*. ICSE-SEIP 2025.
- [5] X. Wang, B. Li, Y. Song et al. *OpenHands: An open platform for AI software developers as generalist agents*. ICLR 2025.

— OBSERVABILITY — END-TO-END TRACING

one user request → spans across the specialist pipeline, with HOTL pauses and repair loops visible as first-class primitives

